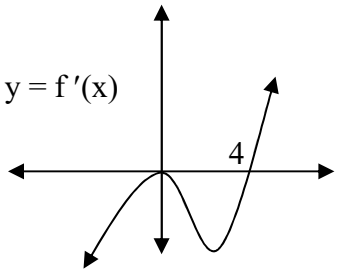
Final Exam Review problems
AP Calculus Name_____

The questions in this packet will help you review for the final exam. Your final exam is cumulative and may include questions not included in this review packet.

Solve the following problems:

1.) Find the limit: $\lim_{h \to 0} \frac{3(x+h)^2 - 3x^2}{h}$

2.) The graph at the right is the graph of the derivative, $f'(x)$, of a function. For what values of x would an accurate graph of f(x) be increasing? decreasing? Use inequality notation.



3.) For what values of x is $f(x) = -5x^2 - 11x + 9$ concave down?

4.) For what values of x is $f(x) = 3x^3 + 5x^2 - 8x + 10$ concave up?

5.) Let $f(x) = \begin{cases} \frac{x^2 - 9}{x - 3} & x \neq 3 \\ 7 & x = 3 \end{cases}$,

a.) What is $\lim_{x \to 3} f(x)$?

b.) What is f(3)?

c.) Is f(x) is continuous at x = 3?

6.) Find $g'(2)$ if $g(x) = \frac{5 + 2x}{x}$.

For exercises 7 – 10, use the information provided about differentiable functions f and g.

	f(x)	g(x)	f ′(x)	g ′(x)
x = 1	4	2	5	1/2
x = 3	1	-4	3/2	-1

7.) Find $(f \cdot g)'(3)$

8.) Find $\left(\frac{f}{g}\right)'(1)$

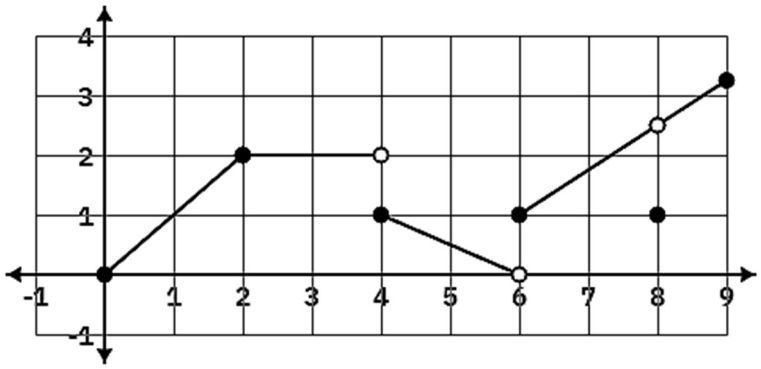
9.) If $J(x) = g(f(x))$, find $J'(3)$

10.) Find $\left(f^{-1}\right)'(4)$

11.) What is the slope of the tangent line to the graph of $x^2 - 3xy^2 + 3y^2 = 1$ at $(-3, 1)$?

12.) A function has derivative $f'(x) = 7x(x + 5)(x - 3)$. For what values of x is the function increasing? For what values of x is the function decreasing?

13.) Given the graph below, what is true about the limits and continuity at the following points: x = 2, x = 4, x = 6, x = 8



MULTIPLE CHOICE PROBLEMS

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I cannot stop you from using your graphing calculator but use it sparingly. Try to do each problem without the calculator. Unless the problem instructions tell you to use a graphing calculator, all these problems can be done without one. You should SHOW WORK FOR EVERY PROBLEM.

Part A:

1. Find all intervals on which the graph of $y=\frac{x^2+1}{x^2}$ is concave upward.

   a. $(-\infty,\infty)$ b. $(-\infty,-1)\cup(1,\infty)$ c. $(-\infty,0)\cup(0,\infty)$ d. $(1,\infty)$

2. If $f(3)=0, f'(3)=6, g(3)=1, g'(3)=\frac{1}{3}$, find $h'(3)$ if $h(x)=\frac{f(x)}{g(x)}$

   a. 18 b. 6 c. –6 d. –2

3. Find all open intervals on which $f(x)=\frac{x}{x^2+x-2}$ is decreasing.

   a. $(-\infty,\infty)$ b. $(-\infty,-2)\cup(-2,1)\cup(1,\infty)$ c. $(-\infty,-2)\cup(1,\infty)$ d. $(-\infty,0)$

4. Find all critical values for $f(x)=(9-x^2)^{3/5}$

   a. 0 b. 3 c. –3, 3 d. –3, 0, 3

5. Find all intervals on which the graph of $f(x)=\frac{x-1}{x+3}$ is concave upward.

   a. $(-\infty,\infty)$ b. $(-\infty,-3)$ c. $(1,\infty)$ d. $(-3,\infty)$

6. Find $\frac{dy}{dx}$ for $5x^2-2xy+7y^2=0$

   a. $\frac{5x+7y}{2}$ b. $\frac{y-5x}{7y-x}$ c. $10x+12y$ d. $\frac{5x-y}{7y-x}$

7. Given $f(x)=10-\frac{16}{x}$, find all c in the interval [2, 8] that satisfies the Mean Value Theorem.

a. 4 b. 5 c. $\frac{8}{5}$ d. ± 4

8. Given $f(x)=\frac{-6}{x}$, choose the correct statement.

a. The graph of f is concave upward on the interval $(-\infty,0)$.
b. The graph of f is concave downward on the interval $(-\infty,0)$.
c. The graph of f is concave upward on the interval $(-\infty,\infty)$.
d. The graph of f is concave upward on the interval $(0,\infty)$.

9. Find all points of extrema on the interval $[0,2\pi]$ if $y=x+\sin x$.

a. $(-\pi,\pi)$ b. (π,π) c. $\left(\frac{3\pi}{2},0\right)$ d. none

10. Which of the following statements is true of $f(x)=-x^3+18x^2-105x+198$?

a. f is decreasing on the interval $(6,\infty)$.
b. f is decreasing on the interval $(5,7)$.
c. f is increasing on the interval $(-\infty,5)$.
d.. f is increasing on the interval $(5,7)$.

12. Find all points of inflection: $f(x)=\frac{1}{12}x^4-2x^2+15$

a. $(2,0)$ b. $(2,0),(-2,0)$ c. $(0,15)$ d. $\left(2,\frac{25}{3}\right)\left(-2,\frac{25}{3}\right)$

13. A function f and its derivative take on the values shown in the table. If g is the inverse of f, find $g'(6)$.

x	$f(x)$	$f'(x)$
2	6	1/4
6	4	2/3

a. $\frac{3}{2}$ b. 4 c. $\frac{8}{3}$ d. undefined

14. A particle is moving along the x-axis according to the position function $s(t)=3t^2-t^3$. What is its maximum velocity?

a. 1 b. 2 c. 3 d. 4

15. Find the absolute maximum and absolute minimum of f on the interval $(0,3]$: $f(x)=\dfrac{x^3-4x^2+7x}{x}$.

a. Maximum: None Minimum: (3, 4)
b. Maximum (0, 7) Minimum: (3, 4)
c. Maximum: None Minimum: (2, 3)
d. Maximum: (0, 7) Minimum: (2, 3)

16. Find $\lim\limits_{x\to 5\pi/4} \dfrac{\tan x-\tan(5\pi/4)}{x-5\pi/4}$

a. 1 b. 2 c. $\sqrt{2}$ d. undefined

18. Find the values of x that give relative extrema for the function $f(x)=(x+1)^2(x-2)$.

a. Relative maximum: $x=-1$; Relative minimum: $x=1$
b. Relative maxima: $x=1$, $x=3$; Relative minimum: $x=-1$
c. Relative maximum: $x=2$
d. Relative maximum: $x=-1$; Relative minimum: $x=2$

19. Find all points of inflection of the function $f(x)=x^4-6x^3$.

a. $(0,0)$ b. $(0,0),\left(\dfrac{9}{2},-\dfrac{2187}{16}\right)$ c. $(3,-81)$ d. $(0,0),(3,-81)$

20. A particle moves along the x-axis with position $x(t)=te^{-t}$. For what value of t is the particle speeding up?

a. $t=1$ b. $t=1.5$ c. $t=2$ d. $t=3$

21. Find $\dfrac{dy}{dx}$ for $y=\sqrt{\sec 4x}$

a. $\dfrac{2}{\sqrt{\sec 4x}}$ b. $\dfrac{\sec 4x\tan 4x}{2\sqrt{\sec 4x}}$ c. $\dfrac{1}{2\sqrt{\sec 4x\tan 4x}}$ d. $2\sqrt{\sec 4x}\tan 4x$

22. Given that $f(x)=-x^2+12x-34$ has a relative maximum at $x=6$, choose the correct statement.

a. f' is positive on the interval $(6,\infty)$.
b. f' is positive on the interval $(-\infty,\infty)$.
c. f' is negative on the interval $(6,\infty)$.
d. f' is negative on the interval $(-\infty,6)$.

23. If f is a function increasing for all values of x, g is a function decreasing for all values of x, describe the behavior of $h(x)=f(g(-x))$.

a. always increasing
b. always decreasing
c. increasing if $x>0$, decreasing if $x<0$
d. decreasing if $x>0$, increasing if $x<0$

25. Given $f(x)=\frac{-1}{x}$, find all c in the interval $\left[-3,-\frac{1}{2}\right]$ that satisfies the Mean Value Theorem.

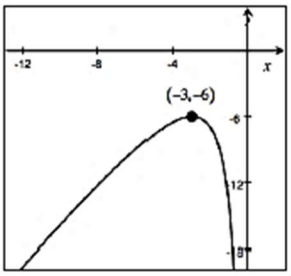
a. $c=-\sqrt{\frac{3}{2}}$

b. $c=\pm\sqrt{\frac{3}{2}}$

c. The Mean Value Theorem doesn't apply because f is not continuous at $x = 0$.

d. The Mean Value Theorem doesn't apply because $f\left(-\frac{1}{2}\right)\neq f(-3)$.

26. On the graph to the right, the point (–3, –6) is critical. Which of the following is true?



I. $f(-3)<0$ II. $f'(-3)<0$ III. $f''(-3)<0$

a. I only
b. I and II only
c. I and III only
d. II and III only

29. Which of the following describes the behavior of $f(x)=x^3-x$.

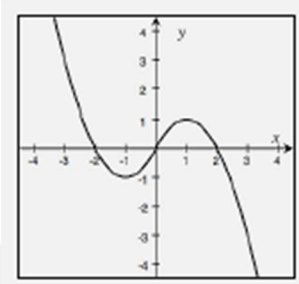
a. Relative maximum: $x = 0$

b. Relative minimum: $x = 0$

c. Relative maximum: $x=\frac{1}{\sqrt{3}}$; relative minimum $x=-\frac{1}{\sqrt{3}}$

d. Relative minimum: $x=\frac{1}{\sqrt{3}}$; relative maximum $x=-\frac{1}{\sqrt{3}}$

30. IF the function to the right is $f(x)$, for what value of x is the product $f(x)f'(x)f''(x)>0$?

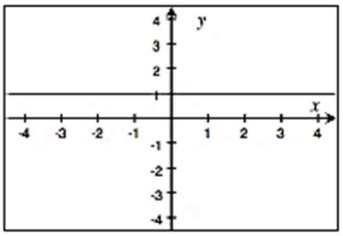


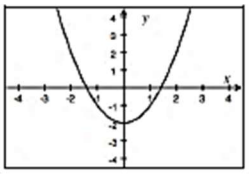
a. 1
b. 1.5
c. –1
d. 3

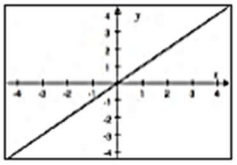
32. If $f(x)=\cos(\ln x)$, describe the behavior of the curve at $x = 1$.

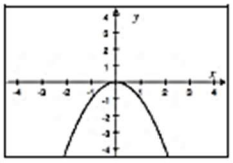
a. relative maximum
b. relative minimum
c. y-intercept
d. inflection point

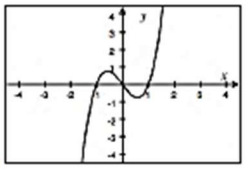
35. The figure in the graph to the right is the 2nd derivative of a polynomial function, f. Choose a possible graph of f.



a. 

b. 

c. 

d. 

36. A particle moves along the x-axis with position function $x(t)=t^2-8t-1$ on the interval [0, 10]. How far does it travel?

a. 18 b. 19 c. 20 d. 52

37. State why Rolle's Theorem does not apply to $f(x)=x^{2/3}-x$ on the interval [–1, 1].

a. f is not continuous on [–1, 1].
b. f is not defined on the entire interval.
c. f is not differentiable at $x=0$.
d. $f(1)\neq f(-1)$.

39. Determine the open intervals where the graph of $f(x)=-\dfrac{1}{(x+1)^2}$ is concave up or concave down.

a. concave down $(-\infty,\infty)$
b. concave down $(-\infty,-1)$; concave up $(-1,\infty)$
c. concave down $(-\infty,-1)$ and $(-1,\infty)$
d. concave up $(-\infty,-1)$ and $(-1,\infty)$

Part B:

Limits

1. $\lim\limits_{x\to -2}\dfrac{-3x-6}{x^2+x-2}$ is

A. –1 B. $\dfrac{-1}{3}$ C. $\dfrac{1}{3}$ D. 1

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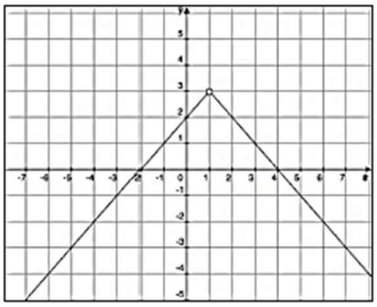
2. The figure to the right shows the graph of $f(x)$. Which of the following statements are true?



I. $\lim\limits_{x\to 1^-} f(x)$ exists

II. $\lim\limits_{x\to 1^+} f(x)$ exists

III. $\lim\limits_{x\to 1} f(x)$ exists

A. I only B. II only C. I and II only D. I, II and III

3. Let $f(x)=\begin{cases} \dfrac{x^2-6x+9}{x-3} & \text{if } x \neq 3 \\ \sqrt{3x} & \text{if } x = 3 \end{cases}$ Which of the following statements I,II, and III are true?

I. $\lim_{x\to 3} f(x)$ exists II. $f(3)$ exists III. f is continuous at $x=3$

A. only I B. only II C. I and II D. none of them

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4. $\lim_{x\to-\infty} \dfrac{-4-x-x^2}{2x^2+3x-2} =$ is

A. 0 B. ∞ C. -2 D. $\dfrac{-1}{2}$

Derivatives

5. If $f(x)=(1-4x^2)^4$, then $f'(x)=$

A. $-32x(1-4x^2)^3$ B. $4(1-8x)^3$ C. $-32(1-8x)^3$ D. $4(1-4x^2)^3$

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6. The slope of the tangent line to $y=\dfrac{x}{2x-1}$ at $x=4$ is

A. $\dfrac{1}{2}$ B. $\dfrac{-1}{14}$ C. $\dfrac{1}{14}$ D. $-\dfrac{1}{49}$

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7. The second derivative of $f(x)=\sin x\cos x$ is

A. 0 B. $-4\sin x\cos x$
C. $(\cos x)^2-(\sin x)^2$ D. $2\cos x-2\sin x$

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8. If $f(-2)=4$ and $f'(-2)=-1$, find the derivative of $\dfrac{f(x)}{x^2}$ at $x=-2$.

A. $\dfrac{3}{4}$ B. 1 C. $\dfrac{1}{4}$ D. $-\dfrac{1}{4}$

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9. Find the equation of the line tangent to $y = \sec x$ at $x = \frac{\pi}{4}$.

A. $y - \frac{\sqrt{2}}{2} = \sqrt{2}\left(x - \frac{\pi}{4}\right)$

B. $y - \sqrt{2} = \sqrt{2}\left(x - \frac{\pi}{4}\right)$

C. $y = \sqrt{2}\left(x - \frac{\pi}{4}\right)$

D. $y - \sqrt{2} = x - \frac{\pi}{4}$

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11. Find the linear approximation to $f\left(\frac{1}{2}\right)$ for $f(x) = e^{\sin x + \cos x}$ at $x = 0$.

Implicit Differentiation

14. If $x^2 + xy - y = 7$, find $\frac{dy}{dx}$ at $(3,-1)$

a. 1 b. $-\frac{5}{2}$ c. -1 d. $\frac{1}{2}$

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15. At which points is the tangent line to the curve $8x^2 + 2y^2 = 6xy + 14$ vertical?

I. $(-2, -3)$ II $(3, 8)$ III. $(4, 6)$

a. I only b. II only c. III only d. I and III only

Continuity and Differentiability

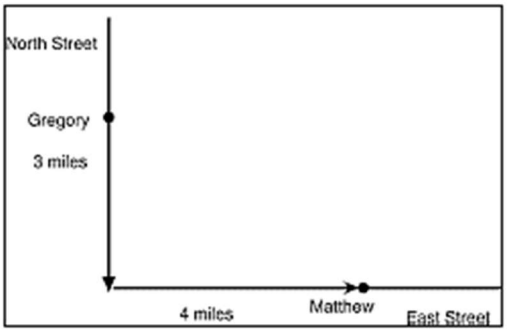
17. If $f(x) = \begin{cases} x^2 - 5x + 8, x \geq 3 \\ x - 1, x < 3 \end{cases}$, describe the behavior of the graph of $f(x)$ at $x = 3$

I. $\lim_{x \to 3} f(x)$ exists II. $f(x)$ is continuous at $x = 3$ III. $f(x)$ is differentiable at $x = 3$

a. I only b. II only c. I and II only d. I, II and III

Related Rates

18. Matthew is visiting Gregory at his home on North Street. Shortly after Matthew leaves, Gregory realizes that Matthew left his wallet and begins to chase him. When Gregory is 3 miles from the 90° intersection along North Street traveling at 45 mph towards the intersection, Matthew is 4 miles along East street traveling away from the intersection at 30 mph. At that time, how fast is the distance between the two men changing?



A. getting closer at 3 mph

B. getting further away at 51 mph

C. getting closer at 51 mph

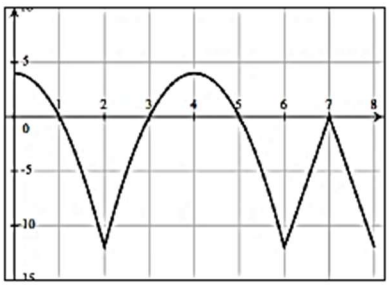
D. Getting closer at 15 mph

19. A cylinder has both its height and radius changing. Its height is increasing at the rate of 3 meters/min. When the height of the cylinder is 8 meters and its radius is 2 meters, the volume is not changing. How fast is the radius decreasing in meters/min? (The volume of a cylinder is given by $V = \pi r^2 h$).

A. 3 B. $\frac{3}{8}$ C. 6 D. 3π

Straight-Line Motion

26. A locomotive is moving along a straight track. Its velocity v of the locomotive at time t, $0 \le t \le 8$ is given by the function whose graph is to the right. At what value of t does the locomotive change direction?



A. 4 only B. 2, 4 and 6 only

C. 1, 3, and 5 only D. 1, 3, 5 and 7 only

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27. A particle moves along a horizontal line with position $x(t) = \frac{10}{t}$. Describe its motion at $t = 1$.

A. Moving right and slowing down B. Moving right and speeding up
C. Moving left and slowing down D. Moving left and speeding up

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Absolute Extrema

29. If $f(x) = \frac{x^3}{3} - x^2$ on $[-2,3]$, $f(x)$ has an absolute minimum at

A. $x = -2$ B. $x = 0$ C. $x = 2$ D. $x = 0$ and $x = 2$

L'Hospital's Rule

32. Find $\lim_{x \to 1} \frac{\sqrt{x} \ln x}{x^2 - 1}$

A. 0 B. $\frac{1}{2}$ C. 1 D. undefined